AWH Series

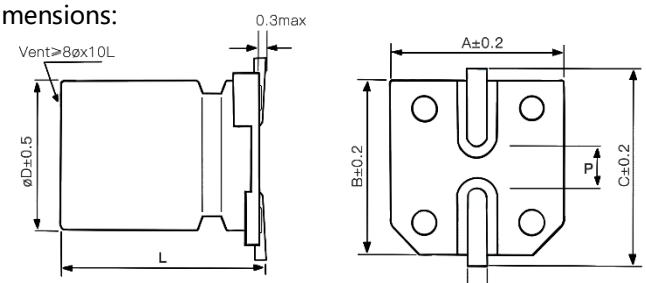
Features

- φ6.3 ~ φ12.5, 125℃, 2000~3000 hours assured
- Low impedance capacitors
- For automobile modules and other high temperature applications
- Designed for reflow soldering
- Designed for surface mounting on high-density PCB
- Vibration resistant structure
- RoHS compliant, REACH&SVHC compliant
- AEC-Q200 compliant, Please contact Jarson for more details, test data, information

Marking color: Black

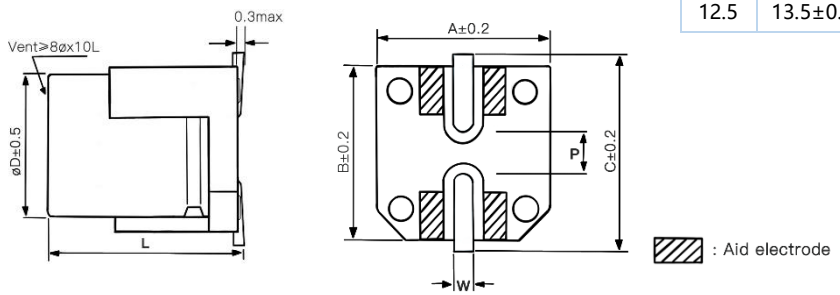
Specifications										
Category temp. range	–40℃ to +125℃									
Capacitance tolerance	±20% (120 Hz / +20 ℃)									
Leakage current	I ≦ 0.01 CV or 3 μA whichever is greater (after 2 minutes)									
Tan δ	Please see the attached characteristics list									
Characteristics at low temperature	Rated voltage (V)	10	16	25	35	50	63	80	100	Impedance ratio at 120 Hz
	Z(-40 ℃)/Z(+20 ℃)	12	8	6	4	4	3	3	3	
Endurance	After applying rated working voltage for 2000/3000 hours at +125 ℃ ± 2 ℃, and then being stabilized at +20 ℃, capacitors shall meet the following limits.									
	Test Time	φD = 6.3mm: 2000H φD = 8~12.5mm: 3000H								
	Capacitance change	Within ±30% of the initial value								
	Dissipation factor (tan δ)	Less than 300% of the initial value								
	Leakage current	Within the initial limit								
Shelf life	After storage for 1000 h at +125 ℃ ± 2 ℃ with no voltage applied and then being stabilized at +20 ℃, capacitors shall meet the limits specified in endurance.									
Resistance to soldering heat	After reflow soldering and then being stabilized at +20 ℃, capacitors shall meet the following limits.									
	Capacitance change	Within ±10% of the initial value								
	Dissipation factor (tan δ)	Within the initial limit								
	Leakage current	Within the initial limit								
Ripple Current and Frequency Multipliers	Frequency	50Hz	120Hz	300Hz	1kHz	10kHz≦				
	Correction Factor	0.35	0.5	0.64	0.83	1.0				

Dimensions:



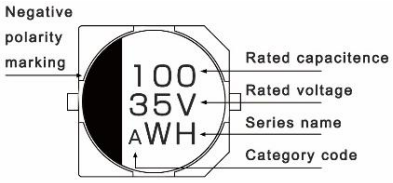
Dimensions						Unit: mm
φD	L	A	B	C	W	P±0.2
6.3	7.7±0.3	6.6	6.6	7.3	0.5~0.8	2.0
8	10.5±0.5	8.3	8.3	9.1	0.7~1.3	3.1
10	10.5±0.5	10.3	10.3	11.1	0.7~1.3	4.4
12.5	13.5±0.5	13.0	13.0	14.0	1.1~1.4	4.4

Vibration resistant structure:

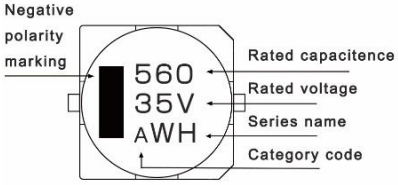


Marking:

ΦD ≦ 10mm



ΦD ≧ 12.5mm



Characteristics list

Rated voltage (V)	Capacitance (±20%) (μF)	Case size øD (mm)	Case size L (mm)	Specification Rated ripple current① (mA rms)	Specification Imp.② (Ω)	Specification tan δ③	Part Number④	Taping&Reel MPQ (pcs/reel)
10	220	8	10.5	270	0.20	0.30	AWH1A221M0810	500
	330	8	10.5	270	0.20	0.30	AWH1A331M0810	500
		10	10.5	500	0.15	0.30	AWH1A331M1010	500
	470	10	10.5	500	0.15	0.30	AWH1A471M1010	500
16	100	6.3	7.7	197	0.50	0.23	AWH1C101M0607	1000
		8	10.5	270	0.20	0.23	AWH1C101M0810	500
	220	8	10.5	270	0.20	0.23	AWH1C221M0810	500
	330	10	10.5	500	0.15	0.23	AWH1C331M1010	500
	470	10	10.5	500	0.15	0.23	AWH1C471M1010	500
25	100	6.3	7.7	197	0.50	0.18	AWH1E101M0607	1000
		8	10.5	270	0.20	0.18	AWH1E101M0810	500
	220	8	10.5	270	0.20	0.18	AWH1E221M0810	500
		10	10.5	500	0.15	0.18	AWH1E221M1010	500
	330	10	10.5	500	0.15	0.18	AWH1E331M1010	500
	470	12.5	13.5	1700	0.08	0.18	AWH1E471M1313	200
	820	12.5	13.5	1700	0.08	0.18	AWH1E821M1313	200
	1000	12.5	13.5	1700	0.08	0.18	AWH1E102M1313	200
35	33	6.3	7.7	197	0.50	0.16	AWH1V330M0607	1000
	47	6.3	7.7	197	0.50	0.16	AWH1V470M0607	1000
		8	10.5	270	0.20	0.16	AWH1V470M0810	500
	100	8	10.5	270	0.20	0.16	AWH1V101M0810	500
	220	10	10.5	500	0.15	0.16	AWH1V221M1010	500
	330	10	10.5	500	0.15	0.16	AWH1V331M1010	500
	470	12.5	13.5	1700	0.08	0.16	AWH1V471M1313	200
	560	12.5	13.5	1700	0.08	0.16	AWH1V561M1313	200
	680	12.5	13.5	1700	0.08	0.16	AWH1V681M1313	200
50	22	6.3	7.7	197	0.50	0.16	AWH1H220M0607	1000
	33	6.3	7.7	197	0.50	0.16	AWH1H330M0607	1000
		8	10.5	270	0.25	0.16	AWH1H330M0810	500
	47	8	10.5	270	0.25	0.16	AWH1H470M0810	500
	82	10	10.5	500	0.20	0.16	AWH1H820M1010	500
	100	10	10.5	500	0.20	0.16	AWH1H101M1010	500
	220	12.5	13.5	1000	0.15	0.16	AWH1H221M1313	200

① Rated ripple current (100kHz / +125℃) ② Impedance (100kHz / +20℃) ③ tan δ (120Hz / +20℃)

※Please refer to the page of reflow conditions for reflow profile.

All product specifications in the catalog are subject to change without notice. Ask factory for the current technical specifications before purchase. (CAT.2026C1)

Characteristics list

Rated voltage (V)	Capacitance (±20%) (μF)	Case size		Specification			Part Number④	Taping&Reel
		øD (mm)	L (mm)	Rated ripple current① (mA rms)	Imp.② (Ω)	tan δ③		MPQ (pcs/reel)
63	10	6.3	7.7	60	2	0.12	AWH1J100M0607	1000
	22	8	10.5	100	0.7	0.12	AWH1J220M0810	500
	33	8	10.5	100	0.7	0.12	AWH1J330M0810	500
		10	10.5	170	0.5	0.12	AWH1J330M1010	500
	47	10	10.5	100	0.7	0.12	AWH1J470M1010	500
		12.5	13.5	170	0.5	0.12	AWH1J470M1313	200
	150	12.5	13.5	1000	0.2	0.12	AWH1J151M1313	200
	180	12.5	13.5	1000	0.2	0.12	AWH1J181M1313	200
	220	12.5	13.5	1000	0.2	0.12	AWH1J221M1313	200
80	10	8	10.5	70	0.75	0.12	AWH1K100M0810	500
	22	8	10.5	70	0.75	0.12	AWH1K220M0810	500
		10	10.5	115	0.55	0.12	AWH1K220M1010	500
	33	8	10.5	70	0.75	0.12	AWH1K330M0810	500
		10	10.5	115	0.55	0.12	AWH1K330M1010	500
	47	10	10.5	115	0.55	0.12	AWH1K470M1010	500
	150	12.5	13.5	700	0.28	0.12	AWH1K151M1313	200
100	10	8	10.5	70	0.75	0.10	AWH2A100M0810	500
	22	8	10.5	70	0.75	0.10	AWH2A220M0810	500
		10	10.5	115	0.55	0.10	AWH2A220M1010	500
	33	10	10.5	115	0.55	0.10	AWH2A330M1010	500
	82	12.5	13.5	700	0.28	0.10	AWH2A820M1313	200

① Rated ripple current (100kHz / +125℃) ② Impedance (100kHz / +20℃) ③ tan δ (120Hz / +20℃)

※Please refer to the page of reflow conditions for reflow profile.

Part Number System:

SMD E-Cap	AWH series	16V	100μF	±20 %	6.3 ϕ x7.7L	
A	**WH**	**1C**	**101**	**M**	**0607**	
Category	Series name	Rated voltage	Capacitance	Tolerance	Case Size	Supplement

Note: For more details, please refer to "Part Number System - SMD Type"